

SHOULD-COST ANALYSIS REPORT

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Bottom-Up Manufacturing Cost Model · Fully Traceable Rate Data

Spur Gear, m3 z38, 40mm bore

Commodity: GEAR · Currency: CNY · FX Rate: 9.0498 to GBP · Operations: 3 · Region: CN

Total Should-Cost	Material	Process	Labour	Margin
¥175.76	73.5%	4.7%	1.2%	7.4%

Model Confidence: **Low** · 3 traced operations · 8 data points auditable

Geometry Provenance — MEASURED (OCCT B-rep kernel)

Volume, weight and every feature were measured from the CAD solid by the Open CASCADE kernel. Measured volume 266.0 cm³; mass for the selected material family 2.088 kg.

Material cost and geometry-derived machining are grounded in the actual solid — not an estimate.

Key Assumptions

Annual volume: 200,000 · Region: CN · Commodity: gear

Alloy / material: 20MnCr5 (Gear Steel) · Net weight: 2.088 kg (measured)

Material utilisation: 100% · Overhead: 12% of factory base · Margin: 8% of subtotal · Operations: 3

¥1 — 8-Bucket Cost Breakdown

Cost Bucket	Amount (CNY)	% of Total	Cost Mix Bar
1. Raw Material	¥129.11	73.5%	
2. Process (Machine)	¥8.28	4.7%	
3. Direct Labour	¥2.11	1.2%	
4. Tooling (amortised)	¥2.90	1.7%	
5. Packaging	¥1.18	0.7%	
6. Logistics	¥2.08	1.2%	
Factory Cost	¥145.65	82.9%	

Cost Bucket	Amount (CNY)	% of Total	Cost Mix Bar
7. Overhead (SG&A)	¥17.09	9.7%	
Subtotal	¥162.74	92.6%	
8. Supplier Margin	¥13.02	7.4%	
TOTAL SHOULD-COST	¥175.76	100.0%	

§2 — Commercial Parameters

Overhead Rate	12.0%	Supplier Margin	8.0%
Packaging / part	¥1.18	Logistics / part	¥2.08
Total Tooling Cost	¥580461.77	Amortisation Volume	200,000 parts

§3 — Material Detail

Parameter	Value	Unit	Notes
Material ID	mat-steel-20mncr5	ID	
Grade / Specification	20MnCr5 (Gear Steel)		Index-anchored 2026-07 (see FORGING PRICING BASIS). European case-hardening gear steel — transmission gears, shafts; excellent hardenability and grain control. Regional adj. ×0.880 (millSteel)
Region	China		
Net Finished Weight	2.0880 kg	kg	Weight in finished part
Direct Material Cost	¥98.25	CNY	Bypasses weight-based model
Data Confidence	Medium		
Effective Date	2026-07		

§4 — Operations Detail

4A Machine Operations

Operation	Machine Class	Rate / hr	Cycle (min)	Parts / Cycle	OEE %	Machine Cost
Gear hobbing	CNC Gear Hobber — small ("d m 4 , "d Ø 2 0 0	¥265.03	0.95	1	80.0%	¥5.23
Chamfer and deburr	Gear Chamfer / Deburr Cell	¥123.89	0.43	1	80.0%	¥1.10
Gear inspection	CNC Gear Checker (profile / lead / pitch)	¥272.86	0.34	1	80.0%	¥1.95
TOTAL						¥8.28

4B Labour Detail

Operation	Labour Grade	Rate / hr	Man ning	Lab. min	Eff %	Labour Cost	Op Total
Gear hobbing	Skilled Machinist	¥71.49	1	0.95	90.0 %	¥1.25	¥6.48
Chamfer and deburr	Semi-skilled Operator	¥49.77	1	0.43	90.0 %	¥0.39	¥1.50
Gear inspection	CMM / Quality Inspector	¥72.40	1	0.34	90.0 %	¥0.46	¥2.41
TOTAL						¥2.11	¥10.39

4C Machined Features — Geometry Audit

Feature	Ø×depth / area	Qty	Operation	Min	Cost	Costed ?
Hole / bore	Ø40.0 × 30 thru	1	Helical mill / bore	2.00	¥8.48	Yes
Face	8865 mm²	2	Face milling (facing)	2.62	¥11.09	Yes
TOTAL costed		3		4.62	¥19.56	Yes

§5 — Machine Rate Buildup

CNC Gear Hobber — small ("dm 4, "d Ø 200) [gear-hob-small] · Confidence: Low · REPRESENTATIVE — UK gear hobber class. Target ~£56/hr. Replace with plant data. | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @80% util	Notes
Depreciation	¥308597.29	¥96.44	4,000 hr/yr available
Maintenance	¥159276.02	¥49.77	
Energy	¥71611.25	¥22.38	
Floor Space	¥54751.13	¥17.11	
Indirect Support	¥149321.27	¥46.66	
Finance Cost	¥104524.89	¥32.66	
TOTAL — CNC Gear Hobber — small ("dm 4, "d Ø 200)	¥848081.84	¥265.03	Effective hours: 3200/yr

Gear Chamfer / Deburr Cell [gear-deburr] · Confidence: Low · REPRESENTATIVE — UK chamfer/deburr cell class. Target ~£26/hr. Replace with plant data. | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @82% util	Notes
Depreciation	¥129411.76	¥39.45	4,000 hr/yr available
Maintenance	¥69683.26	¥21.24	
Energy	¥33051.35	¥10.08	
Floor Space	¥39819.00	¥12.14	
Indirect Support	¥89592.76	¥27.31	
Finance Cost	¥44796.38	¥13.66	
TOTAL — Gear Chamfer / Deburr Cell	¥406354.52	¥123.89	Effective hours: 3280/yr

CNC Gear Checker (profile / lead / pitch) [gear-checker] · Confidence: Low · REPRESENTATIVE — UK CNC gear metrology class (Klingelnberg/Gleason type). Target ~£56/hr. Replace with plant data. | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @65% util	Notes
Depreciation	¥288687.78	¥111.03	4,000 hr/yr available
Maintenance	¥109502.26	¥42.12	
Energy	¥27542.79	¥10.59	
Floor Space	¥59728.51	¥22.97	
Indirect Support	¥149321.27	¥57.43	
Finance Cost	¥74660.63	¥28.72	

How to read the machine rate

Rate/hr = (annual depreciation + maintenance + energy + floor + indirect + finance) ÷ effective hours, where effective hours = available hours × utilisation.

Effective hours encode the shift pattern: a lower hours-base raises £/hr and a higher one lowers it. A challenge from a supplier will target this divisor and whether depreciation is machine-only or full-cell (machine + dosing furnace + trim + robots/extraction) — state your basis when defending the number.

§6 — Rate Traceability

Field	Value	Unit	Source / Reference	Rate ID	Conf.
rawMaterial.directCost	10.8571	£	Pre-computed by commodity module	mat-steel-20mncr5	Medium
rawMaterial.consumablesCostPerPart	3.4090	£	Per-part consumable (core/wax/shell)	mat-steel-20mncr5	Medium
Gear hobbing.machineRatePerHr	29.2853	£/hr	REPRESENTATIVE — UK small CNC hobber class. Target ~£56/hr. Replace with plant data. Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	gear-hob-small	Low
Gear hobbing.labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Chamfer and deburr.machineRatePerHr	13.6897	£/hr	REPRESENTATIVE — UK chamfer/deburr cell class. Target ~£26/hr. Replace with plant data. Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	gear-deburr	Low
Chamfer and deburr.labourRatePerHr	5.5000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-semiskilled	Low
Gear inspection.machineRatePerHr	30.1513	£/hr	REPRESENTATIVE — UK CNC gear metrology class (Klingelnberg/Gleason type). Target ~£56/hr. Replace with plant data. Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	gear-checker	Low
Gear inspection.labourRatePerHr	8.0000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-inspector	Low

§7 — Should-Cost with Uncertainty

Monte-Carlo · Low confidence

Estimate (± band)	P10 · optimistic	P50 · median	P90 · conservative	Band	CV
¥175.76 ± 24.6%	¥135.29	¥171.58	¥221.63	wide	19.7%

Range reflects how well each cost driver is known (High / Medium / Low confidence). Tighten it by attaching CAD, a BOM, or logging an actual quote.

Why confidence is Low

Main band drivers: 6 rate(s) at Low confidence; tooling amortisation carries the widest per-bucket spread.

What's excluded from this unit cost

One-time NRE — tooling / die, fixtures and CNC programming — is amortised separately, not in this unit cost.

Import duty and international freight (regional table is Ex-Works).

Formal embodied-carbon reporting (figures are indicative cradle-to-gate — replace with supplier EPDs).

§8 — Sensitivity Analysis

± 10% driver swing · ranked by ¥ impact

Cost Driver	Base	-10% cost	+10% cost	Range ¥
Direct material cost	¥98.25	-6.8%	+6.8%	¥23.77
Overhead %	12%	-1%	+1%	¥3.69
Supplier Margin %	8%	-0.7%	+0.7%	¥2.60
Gear hobbing: Cycle Time	0.02hr	-0.4%	+0.4%	¥1.57
Gear hobbing: Machine Rate	¥265.03/hr	-0.4%	+0.4%	¥1.26
Amortisation Volume	200000parts	+0.2%	-0.2%	¥0.71
Tooling Cost	¥580461.77	-0.2%	+0.2%	¥0.70
Gear inspection: Cycle Time	0.01hr	-0.2%	+0.2%	¥0.58

§9 — Regional Cost Comparison

Ex-Works · per-region should-cost · vs China

Region	Material	Process	Labour	Tooling	Overhead	Ex-Works	Logistics	Total	vs China
United Kingdom (GBP)	¥146.71	¥15.06	¥7.59	¥5.28	¥22.78	¥197.41	¥1.44	¥213.55	+22%
Germany (EUR)	¥151.11	¥15.81	¥12.26	¥5.54	¥26.51	¥211.23	¥1.65	¥228.59	+30%
France (EUR)	¥149.64	¥13.85	¥8.81	¥4.85	¥24.27	¥201.43	¥1.65	¥218.15	+24%
Spain (EUR)	¥146.71	¥13.25	¥5.56	¥4.64	¥21.09	¥191.25	¥1.65	¥207.25	+18%
Poland (PLN)	¥142.31	¥10.84	¥3.45	¥3.80	¥17.79	¥178.19	¥1.72	¥193.20	+10%
Turkey (TRY)	¥132.04	¥9.03	¥1.92	¥3.17	¥14.87	¥161.03	¥1.87	¥174.90	-0%
China (CNY)	¥129.11	¥8.28	¥2.11	¥2.90	¥17.09	¥159.48	¥2.08	¥175.76	Base
India (INR)	¥132.04	¥7.83	¥1.34	¥2.74	¥13.52	¥157.48	¥2.15	¥171.16	-3%
Mexico (MXN)	¥139.37	¥9.03	¥2.22	¥3.17	¥15.65	¥169.45	¥1.94	¥183.86	+5%
United States (USD)	¥146.71	¥12.80	¥9.96	¥4.49	¥21.56	¥195.52	¥1.79	¥211.82	+21%

Per-region should-cost using regional labour, energy and rate benchmarks. Tooling held fixed. Ex-Works — excludes import duty + international freight. Indicative — confirm with RFQ.

§10 — Embodied Carbon

8.85 kgCO2e · 4.24 /kg · cradle-to-gate

Material	4.38 kgCO2e	Material factor	2.10 kg/kg (Steel)
Process	4.13 kgCO2e	Grid intensity	0.550 kg/kWh · 7.52 kWh
Logistics	0.33 kgCO2e	Total	8.85 kgCO2e

Indicative cradle-to-gate factors — replace with supplier EPDs for formal reporting.

[CRITICAL] Material is the dominant cost driver

Finding	Raw material at 73.5% of total exceeds the gear benchmark ceiling of 50%. Material is controlling this part's economics.
Impact	High · up to 9% potential saving
Benchmark	Material %: yours 73.5% vs industry 28–50%
Action 1	Material utilisation at 100% — every +5% saves ~£0.71/part
Action 2	Re-cost the bought-out content at the target volume

§12 — Cost-Reduction Opportunities (ranked by saving)

7 opportunities · combined ¥18.63/part (~11%)

Ranked on money per part against this costing. The combined figure is the root-sum-square of the top three, capped at 40% — overlapping actions do not save twice, so the column is deliberately not summed.

#	Category	Action	Why it is on the list	Save/p art	%	Timeframe	Risk
1	Process & Automation	Focus cycle-time engineering on the governing operation before touching anything else; ask the supplier for its OEE and changeover data	One operation dominates — "Gear hobbing" — "Gear hobbing" carries 62% of the conversion cost across 3 operations. The part's cost rides on a single process step.	¥14.06	8.0%	Medium Term	Medium
2	Process & Automation	OEE Improvement Programme (TPM)	OEE at 80% vs 85% world-class target. Each 5% OEE gain reduces effective machine cost by 5%.	¥2.64	1.5%	Medium Term	Low
3	Design & Geometry	Design Review for DFM Compliance	Formal DFM review with manufacturing engineering to identify part features that increase cost without functional benefit.	¥8.79	5.0%	Quick Win	Low
4	Design & Geometry	Fix the burr at source (tooling discipline, gate redesign); where finishing must remain, tumble/vibratory beats manual	Manual finishing after the primary process — Deburring/fettling/rework appears as its own operation — a symptom of an upstream condition (tool wear, gating, die state) being paid for by hand, every part.	¥5.27	3.0%	Quick Win	Low
5	Material	Consumables Rationalisation (Cores, Patterns, Shell)	Per-part consumables are 24% of the material line. Core count, pattern life and shell recipe are all design-controllable.	¥8.40	4.8%	Medium Term	Medium
6	Commercial & Sourcing	Raw-Material Price Indexation Clause	With material at 73.5% of cost, the quote embeds the supplier's hedge against metal/resin volatility. An index-linked clause removes that risk premium from the piece price.	¥5.27	3.0%	Quick Win	Low
7	Commercial & Sourcing	Annual Volume Re-Commitment for Better Pricing	Provide 12-month rolling volume forecast to supplier to secure better unit pricing and tooling amortisation.	¥5.27	3.0%	Quick Win	Low

§13 — Opportunity by Category

4 categories with an actionable lever

Category	Opportunities	Best single action	Best save/part	Category total (indicative)
Process & Automation	2	Focus cycle-time engineering on the governing operation before touching anything else; ask the supplier for its OEE and changeover data	¥14.06	¥16.70
Design & Geometry	2	Design Review for DFM Compliance	¥8.79	¥14.06
Material	1	Consumables Rationalisation (Cores, Patterns, Shell)	¥8.40	¥8.40
Commercial & Sourcing	2	Raw-Material Price Indexation Clause	¥5.27	¥10.55

§15 — Implementation Roadmap

Quick Wins — Implement Immediately

- Raw-Material Price Indexation Clause
- Design Review for DFM Compliance
- Annual Volume Re-Commitment for Better Pricing